

Multi-Step Proximal Policy Improvement: Separating Target-Policy Displacement from Deployment Reach in Offline Actor–Critic Learning

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Abstract

Behavior-regularized offline actor–critic methods commonly use one policy branch both to construct bootstrapped critic targets and to form the deployed policy. This ties the target-policy displacement seen during critic training to the total reach of policy improvement. We introduce multi-step proximal policy improvement (MPI), a re-centered K -hop refinement of TD3+BC in which a target copy of the first hop supplies TD actions and the final hop supplies the evaluated policy. For deterministic actors, squared action anchoring defines a Wasserstein–2 proximal step. We derive a finite-step margin relative to the learned critic, an explicit critic-error transfer term, a first-hop target-perturbation bound, and a fixed- K small-budget expansion. In a complete sweep over nine D4RL locomotion tasks and fourteen fixed budgets, three proximal hops reduce large-budget low-return cells from 35/63 to 17/63; three scale-matched linearized hops reduce them to 24/63. Across audited checkpoints, three-hop MPI lowers deterministic target-policy displacement relative to TD3+BC in 88/90 paired runs, while its current-state final-policy displacement proxy is lower in only 36/90. In a paired shadow-critic intervention on eight targeted runs, the final-route shadow has greater error than the first-route shadow relative to a common simulator return under the first-route continuation in all eight. Together with the failure diagnostics, these results show that the implemented fixed-budget refinement bundle broadens the empirical stability envelope while keeping the theoretical guarantee explicitly relative to the learned critic.

1 INTRODUCTION

Offline reinforcement learning (RL) seeks a policy from a fixed dataset without additional environment interaction. Its central difficulty is not merely to select a high-value action, but to determine how far policy improvement can proceed before the learned critic becomes unreliable. TD3+BC regularizes a deterministic actor toward dataset actions while using a Polyak copy of that actor in bootstrapped critic targets (Fujimoto and Gu, 2021). The same policy branch therefore determines both the action shift entering critic training

and the reach of the evaluated policy. Aggressive improvement can move target actions into regions where critic error is amplified, whereas conservative improvement can leave useful policy improvement unrealized.

We study this coupling through multi-step proximal policy improvement (MPI), a re-centered refinement architecture for TD3+BC. MPI represents a nominal improvement budget with K sequential actor updates. The first actor is anchored to dataset actions, and each later actor is anchored to the stop-gradient output of its predecessor. A target copy of the first refined actor supplies the next actions in TD targets, while the final actor is used for evaluation. The architecture thus assigns critic-target construction and deployment to distinct policy branches without changing the critic update rule.

Let D_{critic} denote the squared displacement of the critic-coupled first-hop branch and let D_{final} denote that of the evaluated final policy. A one-hop actor identifies these quantities; MPI assigns them to μ_1 and μ_K , respectively. This separation can increase final-policy reach at comparable first-hop movement or contract both movements near an unstable training outcome. It also introduces a countervailing mismatch because the critic target follows the μ_1 branch while evaluation follows μ_K . The method therefore exposes a design axis between target-policy displacement and deployment reach rather than guaranteeing that either quantity is independently optimal.

For deterministic actors, the squared action anchor in TD3+BC is a Wasserstein–2 movement cost in the Dirac-policy limit. Under a fixed critic, inactive action projection, accurate subproblem solutions, and small realized steps, explicit and implicit updates are first-order discretizations of the same Wasserstein gradient flow. At finite steps, a proximal comparison yields an improvement margin only relative to the learned critic. At large nominal budgets, projection, neural actor fitting, critic drift, saturation, and the seeded optimization trajectory determine the observed end-to-end outcome. We keep these local, finite-step, and coupled regimes distinct throughout the analysis.

We evaluate this formulation with a complete fixed-budget sweep over nine D4RL locomotion tasks, fourteen budgets, and two paired seeds. Three proximal hops reduce large-budget low-return cells from 35/63 for TD3+BC to 17/63 and primarily improve

the lower tail rather than uniformly shifting scores. Three scale-matched projected linearized hops reduce failures to 24/63, so the stability effect is not unique to the proximal loss. Direct next-state diagnostics show that three-hop MPI lowers deterministic target-policy displacement in 88/90 paired comparisons with TD3+BC, while its current-state final-policy displacement proxy is lower in only 36/90. Low-return runs additionally exhibit orders-of-magnitude increases in TD-error tails, larger realized movements, and critic gains that reverse sign under a controlled common critic. In a paired shadow-critic intervention, the final-route shadow has greater error than the first-route shadow relative to a common simulator return under the first-route continuation in all eight targeted runs.

Our contributions are as follows.

- We formulate MPI for TD3+BC as a re-centered actor composition that assigns target-policy displacement and final-policy reach to distinct branches.
- We give a Wasserstein proximal interpretation, derive an inexact finite-step margin with an explicit critic-error term, bound first-hop TD-target perturbation, and separate these statements from the fixed-critic small-step flow limit.
- Through complete proximal and scale-matched linearized sweeps, direct target-policy measurements, and paired shadow-critic diagnostics, we show that substepping broadens the empirical large-budget stability envelope and that low-return outcomes coincide with critic miscalibration rather than local truncation error alone.

2 BACKGROUND

2.1 Offline actor-critic learning

We consider a discounted Markov decision process $\langle \mathcal{S}, \mathcal{A}, P, r, \gamma \rangle$ with continuous action space $\mathcal{A} \subset \mathbb{R}^d$. Offline RL receives only a fixed dataset $\mathcal{D} = \{(s_i, a_i, r_i, s'_i)\}_{i=1}^N$ and must learn a policy without additional data collection. A critic $\bar{Q}$ is trained from bootstrapped targets of the form

$$y_\rho = r + \gamma \min_j Q_j^-(s', \rho(s')), \quad (1)$$

where ρ is the target policy and Q_j^- are target critics. When ρ selects actions outside dataset support, errors in Q_j^- enter both the regression target and the next actor update. This feedback is a central source of offline instability (Fujimoto et al., 2019; Kumar et al., 2019).

Behavior-anchored algorithms control this feedback by optimizing a value term together with a cloning or divergence penalty. For a deterministic actor μ_θ , the normalized TD3+BC form is

$$\mathcal{L}_{\text{actor}}(\theta) = -\alpha \mathbb{E}_s \bar{Q}(s, \mu_\theta(s)) + \mathbb{E}_{(s,a) \sim \mathcal{D}} \frac{1}{d} \|\mu_\theta(s) - a\|_2^2, \quad (2)$$

where $\bar{Q} = \hat{Q}/C$ and $C > 0$ is a stop-gradient critic scale. The anchor makes the update conservative, but a one-step method uses the same policy branch for both roles: a Polyak target copy enters the TD target, while the online actor is deployed. Consequently, the displacement that perturbs bootstrapping is tied to the total displacement used for policy improvement.

2.2 Proximal updates and Wasserstein geometry

Fix a state distribution ρ and a policy family $\mathcal{M}$. Given an estimated energy $\hat{\mathcal{E}}$ and a metric $d_{\mathcal{M}}$, a proximal policy update is

$$\pi^+ \in \arg \min_{\pi \in \mathcal{M}} \left\{ \hat{\mathcal{E}}(\pi) + \frac{1}{2h} d_{\mathcal{M}}^2(\pi, \pi^{\text{ref}}) \right\}. \quad (3)$$

This is the minimizing-movement, or implicit-Euler, construction associated with a metric gradient flow (Ambrosio et al., 2005; Otto, 2001). The variational problem in Eq. (3) is meaningful for finite h whenever a minimizer exists. Its interpretation as an accurate time step of a continuous flow is more restrictive: it requires a fixed energy, sufficiently small realized moves, and increasingly accurate solves as the step size vanishes.

For state-conditioned deterministic policies $\pi_\mu(\cdot | s) = \delta_{\mu(s)}$, the quadratic Wasserstein distance has the Dirac limit

$$W_2^2(\pi_\mu(\cdot | s), \pi_\nu(\cdot | s)) = \|\mu(s) - \nu(s)\|_2^2. \quad (4)$$

Thus squared action anchoring is a Wasserstein movement cost, up to a fixed metric scaling. We use this observation to separate three levels of analysis: (i) a local Wasserstein-flow limit, (ii) finite-step inexact proximal improvement relative to the learned critic, and (iii) the coupled actor-critic branch dynamics that dominate at large policy-improvement budgets.

3 RELATED WORK

Offline policy regularization. Offline RL methods control distribution shift either by constraining policy improvement or by modifying the learned value function. Behavior constraints, divergence penalties, and squared action anchoring keep the learned policy near dataset support (Fujimoto et al., 2019; Kumar et al., 2019; Wu et al., 2019; Fujimoto and Gu, 2021; Tarasov et al., 2023). Other approaches regularize policy extraction through value-dependent weighting or reshape the critic through pessimism and ensemble diversity (Nair et al., 2020; Kostrikov et al., 2022; Kumar et al., 2020; An et al., 2021). MPI addresses a different control variable within TD3+BC: the movement of the actor branch whose target copy enters critic learning.

Policy evaluation, target lag, and decoupled extraction. Approximate policy iteration conventionally alternates policy evaluation and improvement, but their

learned components can still share a target-policy role. One-step offline RL instead freezes an estimate of the behavior policy’s value before a single constrained improvement (Brandfonbrener et al., 2021). Backup-side constraints instead restrict the actions selected inside the Bellman operator (Kumar et al., 2019; Ghasemipour et al., 2021), while decoupled extraction can train a behavior model independently and use critic guidance only at inference (Lin et al., 2026). Polyak averaging delays a policy copy but retains the same improvement branch for target construction and evaluation. MPI makes a narrower distinction inside TD3+BC training: the first local refinement supplies the target-policy branch and later refinements supply the evaluated branch.

Trust regions, proximal methods, and policy geometry. Natural policy gradient and trust-region methods connect policy optimization to Fisher information geometry (Amari and Nagaoka, 2000; Schulman et al., 2015). Optimal-transport geometry provides a distinct notion of policy displacement: policy optimization has been formulated as a Wasserstein gradient flow, and Wasserstein natural-gradient or trust-region methods have been developed for continuous control (Zhang et al., 2018; Moskovitz et al., 2020; Terpin et al., 2022; Song et al., 2023). In offline RL, Wasserstein discrepancies have mainly been used as behavior regularizers or distribution-matching objectives (Wu et al., 2019; Asadulaev et al., 2024). Our use is different. The distance defines an estimated-energy proximal operator; composing that operator with a moving center defines a refinement path. We also distinguish the finite-step variational statement from the small-step flow limit, rather than treating large-budget end-to-end behavior as a direct numerical-integration test.

Expressive policy extraction and iterative refinement. Flow- and diffusion-based actors improve the expressivity of offline policy extraction (Park et al., 2025; Wang et al., 2023). Their expressivity does not by itself determine how aggressively a learned critic should move the policy, and a single critic-guided extraction step can still be sensitive to off-support value errors. MPI instead isolates sequential re-centering and the policy branch used in critic feedback. The controlled experiments use deterministic actors, for which the Wasserstein movement cost reduces to squared action displacement.

Bootstrapping error and actor-critic coupling. Prior work has emphasized extrapolation and bootstrapping error as defining failure modes of offline value learning (Fujimoto et al., 2019; Kumar et al., 2019, 2020). Our analysis focuses on an actor-side control variable that is usually implicit: the displacement of the policy entering the TD target. The resulting target-perturbation bound formalizes why target-policy dis-

placement and final deployment reach are distinct quantities, while the experiments measure their realized counterparts under a jointly learned critic.

4 MULTI-STEP PROXIMAL POLICY IMPROVEMENT

At each delayed actor update, MPI maintains K independently initialized, persistent deterministic actors with separate optimizer states. The first actor takes one TD3+BC update toward the sampled dataset actions; each later actor takes one update toward the stop-gradient output of its predecessor on the same minibatch. The online critic is fixed within this chain, and its first member is $\hat{Q}$. A Polyak copy of the first actor supplies TD-target actions, and the final actor is evaluated. We first derive the proximal operator underlying these updates and then formalize their composition.

4.1 Offline actor updates as proximal improvement

Let $\rho_{\mathcal{D}}$ be the empirical state distribution and consider a deterministic actor $\mu : \mathcal{S} \rightarrow \mathcal{A} \subset \mathbb{R}^d$. At one training iteration, the critic parameters are held fixed while the actor chain is updated. We write the current critic as $\hat{Q}$ and normalize its global scale by

$$\bar{Q}_k(s, a) = \frac{\hat{Q}(s, a)}{C_k}, \quad (5)$$

$$C_k = \text{sg}\left(\mathbb{E}_s[\hat{Q}(s, \tilde{\mu}_k(s))]\right) + \epsilon.$$

where $\tilde{\mu}_k$ is the actor at the beginning of the k th update and sg denotes stop-gradient. For fixed positive C , normalized and raw-critic ascent have the same integral curves after a constant change of time. Re-computing a positive global scalar C_k preserves the instantaneous raw-critic ascent direction and gives a trajectory-dependent time reparameterization in the infinitesimal limit; at finite steps it changes the effective step schedule. All finite-step statements condition on realized C_k .

For deterministic policies and a state distribution ρ , define the action-averaged Wasserstein metric

$$\mathbf{d}_{d,\rho}^2(\mu, \nu) := \mathbb{E}_{s \sim \rho} \left[\frac{1}{d} \|\mu(s) - \nu(s)\|_2^2 \right]. \quad (6)$$

It is d^{-1} times the state-averaged squared W_2 distance between state-conditioned Dirac policies. We abbreviate $\mathbf{d}_d = \mathbf{d}_{d,\rho_{\mathcal{D}}}$ on the actor-training state distribution. For diagonal-Gaussian policies, the statewise counterpart is $W_2^2 = \|m - m^{\text{ref}}\|_2^2 + \|\sigma - \sigma^{\text{ref}}\|_2^2$, so the same construction acts on mean and standard-deviation coordinates. The deterministic case is its zero-variance limit; the formal statements and fixed-budget mechanism study below use deterministic actors.

The population counterpart of squared behavior cloning also admits a deterministic reference. Let P_B

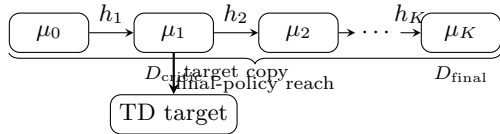


Figure 1: MPI re-centers each actor update. A Polyak copy of μ_1 supplies TD-target actions, while later actors increase or reshape final-policy reach without entering the target policy.

be the data-generating behavior distribution, let ρ_B be its state marginal, and define $b(s) = \mathbb{E}_{\rho_B}[a \mid s]$. Then

$$\mathbb{E}_{(s,a) \sim \rho_B} \frac{1}{d} \|\mu(s) - a\|_2^2 = d_{d,\rho_B}^2(\mu, b) + \text{const}, \quad (7)$$

where the constant is the conditional action variance and does not depend on μ . The empirical actor loss is a Monte Carlo estimate of the left-hand side, so samplewise squared anchoring targets the conditional-mean behavior map at the population level.

Given a reference policy ν and step $h > 0$, define single-step proximal improvement (SPI) by

$$\text{SPI}_h^{\hat{Q}}(\nu) \in \arg \min_{\mu \in \mathcal{M}} \left\{ -\mathbb{E}_s \bar{Q}_k(s, \mu(s)) + \frac{1}{2h} d_d^2(\mu, \nu) \right\}. \quad (8)$$

Multiplying by $2h$ gives the practical actor objective

$$\mathcal{L}_{h,k}(\mu \mid \nu) = -2h \mathbb{E}_s \bar{Q}_k(s, \mu(s)) + d_d^2(\mu, \nu). \quad (9)$$

Proposition 1 (Behavior anchoring is Wasserstein proximal improvement). For deterministic policies, the normalized anchored objective

$$-\alpha \mathbb{E}_s \bar{Q}(s, \mu(s)) + d_d^2(\mu, \nu)$$

is Eq. (8), up to multiplication by a positive constant, with $h = \alpha/2$. Equivalently, the raw-critic objective $-\lambda \mathbb{E}_s \hat{Q}(s, \mu(s)) + d_d^2(\mu, \nu)$ has $h = \lambda C/2$ when C is fixed. The population counterpart of squared dataset anchoring uses $\nu = b$ by Eq. (7).

The proposition identifies an update operator relative to the estimated critic; true-return improvement lies outside the scope of this certificate.

4.2 Re-centered composition and actor-critic dataflow

For population analysis, let $\mu_0 := b$ denote the conditional-mean behavior map and let $h_{1:K}$ be positive step sizes. MPI constructs

$$\mu_k \in \text{SPI}_{h_k}^{\hat{Q}}(\text{sg } \mu_{k-1}), \quad k = 1, \dots, K. \quad (10)$$

The implemented first actor μ_1 uses sampled dataset actions as its reference. Later actors $\mu_2, \dots, \mu_K$ use the preceding online actor under the same current critic. Only the Polyak copy of μ_1 enters the TD target, while μ_K supplies the evaluated action.

Our fixed-budget schedule sets $h_k = T/K$ and holds the total nominal budget T fixed. Increasing K jointly

changes local step size, re-centering frequency, actor compute, and the target/deployment dataflow; the experiment therefore estimates the effect of this implemented refinement bundle.

On the next-state distribution ρ'_D used by the TD target, define the action-averaged population displacements

$$\begin{aligned} D_{\text{critic}} &:= d_{d,\rho'_D}^2(\mu_1, \mu_0), \\ D_{\text{final}} &:= d_{d,\rho'_D}^2(\mu_K, \mu_0). \end{aligned} \quad (11)$$

A one-step actor has $D_{\text{critic}} = D_{\text{final}}$; MPI assigns the two population quantities to distinct policy branches. On a fixed current-state batch, we define the sample-anchored online-actor proxies

$$\begin{aligned} \hat{D}_{\text{critic}} &:= \mathbb{E}_i \frac{1}{d} \|\mu_1(s_i) - a_i\|_2^2, \\ \hat{D}_{\text{final}} &:= \mathbb{E}_i \frac{1}{d} \|\mu_K(s_i) - a_i\|_2^2. \end{aligned} \quad (12)$$

The direct audit reproduces the training transition loader and retains only rows with a nonzero bootstrap mask; timeout boundaries and terminal source rows are excluded. On these rows, we also measure the deterministic target-policy displacement

$$\hat{D}_{\text{target}} := \mathbb{E}_i \frac{1}{d} \|\mu_1^-(s'_i) - a_{i+1}\|_2^2, \quad (13)$$

where μ_1^- is the Polyak target actor and a_{i+1} is the paired next dataset action. A smoothed counterpart uses the clipped TD3 target action with common noise draws. Sample anchoring adds the same conditional-action-variance floor in population comparisons; paired differences cancel this floor. These quantities measure policy displacement rather than value-target perturbation or critic reliability.

4.3 Finite-step statements

The following statements do not require a small-step Taylor expansion.

Proposition 2 (Inexact estimated-energy margin). Suppose μ_{k-1} is feasible for the k th actor class and the computed actor satisfies

$$\mathcal{L}_{h_k,k}(\mu_k \mid \mu_{k-1}) \leq \mathcal{L}_{h_k,k}(\mu_{k-1} \mid \mu_{k-1}) + \varepsilon_k.$$

Then

$$\begin{aligned} m_k^{\hat{Q}} &:= \frac{2h_k}{C_k} \mathbb{E}_s [\hat{Q}(s, \mu_k(s)) - \hat{Q}(s, \mu_{k-1}(s))] \\ &\quad - d_d^2(\mu_k, \mu_{k-1}) \geq -\varepsilon_k. \end{aligned} \quad (14)$$

For the first sample-anchored hop, a statewise dataset action is not generally a feasible deterministic-policy comparator; the population reference in Eq. (7) or a behavior actor is required for a literal certificate. For re-centered hops, the preceding actor is feasible by construction.

The margin is relative to $\hat{Q}$. Let a trusted objective be $Q^\dagger$ and write $\hat{Q} = Q^\dagger + e$. Then

$$m_k^{Q^\dagger} = m_k^{\hat{Q}} - \frac{2h_k}{C_k} \mathbb{E}_s[e(s, \mu_k(s)) - e(s, \mu_{k-1}(s))]. \quad (15)$$

Thus estimated-energy descent transfers to $Q^\dagger$ only when critic error is controlled along the realized path. Equation (14) certifies the learned objective rather than monotone true return.

Proposition 3 (First-hop TD-target perturbation bound). For clarity, consider a noiseless TD target using μ_1 , $y_{\mu_1} = r + \gamma \min_j Q_j^-(s', \mu_1(s'))$, and suppose each target critic $Q_j^-(s', \cdot)$ is $L(s')$ -Lipschitz. For any reference μ_0 ,

$$\|y_{\mu_1} - y_{\mu_0}\|_{L_2(\mathcal{D})} \leq \gamma \left(\mathbb{E}_{s'} L(s')^2 \cdot \|\mu_1(s') - \mu_0(s')\|_2^2 \right)^{1/2}. \quad (16)$$

If $L(s') \leq \bar{L}$, the right-hand side is at most $\gamma \bar{L} \sqrt{dD_{\text{critic}}}$. Conditional on μ_1 , later actors do not enter this one-step target perturbation.

Proposition 3 bounds a one-step value-target difference, not the stability of the learned critic. Its architectural content is that later refinement can alter D_{final} without directly entering this bound. With a Polyak target network, μ_1 is replaced by its target copy. A same-noise comparison under TD3 target smoothing obeys the same bound because action clipping is non-expansive. The current-state $\hat{D}_{\text{critic}}$ omits Polyak lag and target noise, whereas $\hat{D}_{\text{target}}$ measures the Polyak actor directly and has a smoothed counterpart. Neither displacement measures the action Lipschitz factor or the resulting change in critic value.

The separation introduces a complementary evaluation-policy mismatch: critic targets follow the μ_1 branch, whereas evaluation follows μ_K . Increasing D_{final} at fixed D_{critic} can therefore make the critic less representative of the evaluated policy. We interpret MPI as exposing a measurable target/deployment trade-off; its empirical return captures the net effect of target movement, evaluation mismatch, actor compute, and the coupled optimization trajectory.

4.4 Local Wasserstein-flow limit

The gradient-flow interpretation is a local organizing principle. Fix a critic and an interior action a_0 . Let $g = \nabla_a \hat{Q}(a_0)$, $H = \nabla_a^2 \hat{Q}(a_0)$, and $\beta = dh/C$. Before projection, the scale-matched linearized and exact statewise proximal updates satisfy

$$\Delta a_{\text{lin}} = \beta g, \quad \Delta a_{\text{prox}} = \beta \nabla_a \hat{Q}(a_0 + \Delta a_{\text{prox}}). \quad (17)$$

When $\beta \|H\| \ll 1$, the latter yields

$$\Delta a_{\text{prox}} = (I - \beta H)^{-1} \beta g = \beta g + \beta^2 H g + O(\beta^3), \quad (18)$$

and both updates are first-order discretizations of $\dot{a} = d\nabla_a \hat{Q}(a)$. With fixed C this is a constant time-rescaling of raw-critic gradient ascent; with a positive policy-dependent global C it follows the same direction under an adaptive time parameterization. This expansion additionally requires small realized displacement, inactive projection, and a sufficiently accurate actor solve. At large budgets, our experiments diagnose the resulting nonlocal actor-critic regime instead of extrapolating Eq. (18). The end-to-end return sweep, even at small nominal T , is not a frozen-critic action-level consistency test of this limit.

Proposition 4 (Local fixed-budget composition). Let a smooth one-step update in local coordinates satisfy $\Psi_h(x) = x + hf(x) + h^2 p(x) + O(h^3)$. For fixed K and $T \rightarrow 0$,

$$\begin{aligned} \Psi_{T/K}^K(x_0) &= x_0 + T f_0 \\ &+ T^2 \left[\frac{K-1}{2K} J_0 f_0 + \frac{1}{K} p_0 \right] + O(T^3). \end{aligned} \quad (19)$$

where $f_0 = f(x_0)$ and $J_0 = Df(x_0)$. Coordinate explicit Euler has $p_0 = 0$, whereas exact backward Euler has $p_0 = J_0 f_0$. Hence both are first-order methods; for each fixed K , the $T \rightarrow 0$ expansion has a leading flow-bias coefficient proportional to $1/K$.

Proposition 4 explains a possible fixed-critic small-step benefit of substepping. It neither predicts the sign of true-return change nor applies to the projected, saturated, moving-critic branches studied at large T . The expansion is not a uniform $K \rightarrow \infty$ result.

5 EXPERIMENTS

We evaluate MPI on the nine D4RL locomotion tasks formed by HalfCheetah, Hopper, and Walker2d with medium, medium-replay, and medium-expert datasets (Fu et al., 2020). Base implementations follow CORL configurations (Tarasov et al., 2022); scores are D4RL normalized returns. Additional per-task results and diagnostic definitions are in Appendix B.

Fixed-budget protocol and estimand. We hold a total nominal budget T fixed, split it across K actor hops as $h = T/K$, and sweep

$$T \in \{0.05, 0.1, 0.2, 0.4, 0.7, 1.5, 2.5, 4, 7, 10, 12, 14, 17, 20\}.$$

over all nine tasks with two paired seeds. The sweep is complete over this specified grid and contains $9 \times 14 \times 2$ runs per method. It is designed as a high-resolution stability phase diagram rather than a final benchmark alone. For $K = 1$, Proposition 1 maps T to the normalized TD3+BC coefficient $\alpha = 2T$.

We compare TD3+BC ($K = 1$), proximal MPI with $K \in \{2, 3\}$ (MPI-Prox), and a scale-matched projected

Table 1: Aggregate fixed-budget results. High denotes $T \geq 4$ and tail denotes $T \geq 10$. Collapse entries are seed-mean cells / raw seeded runs.

Method	$T \leq 2.5$	High	Tail	Collapse
TD3+BC	68.35	25.97	20.46	35 / 68
MPI-Prox, $K = 2$	71.62	42.02	31.49	25 / 53
MPI-Prox, $K = 3$	70.56	52.78	43.55	17 / 41
MPI-Lin, $K = 2$	69.02	36.68	27.44	29 / 58
MPI-Lin, $K = 3$	68.54	43.91	33.12	24 / 53

linearized target with $K \in \{2, 3\}$ (MPI-Lin):

$$a_{k+1}^{\text{tar}} = \Pi_{\mathcal{A}} \left[a_k + d \frac{T}{K} \frac{\nabla_a \hat{Q}(s, a_k)}{C_k} \right]. \quad (20)$$

Here a_0 is the sampled dataset action. At each hop, MPI-Lin detaches the current reference, recomputes the $\hat{Q}$ gradient and C_k , clips the target to the action box, and takes one mean-squared-error optimizer step with the corresponding persistent actor. It uses the same first-hop target and final-hop evaluation routing as MPI-Prox. The factor d matches the local action-space coefficient induced by the action-averaged metric in Eq. (6). All conditions match datasets, critic updates, and the horizon of 10^6 critic updates; checkpoint diagnostics use the final checkpoint. A K -hop condition performs K actor optimizer steps at each delayed policy update, and C_k is recomputed at each reference. The primary estimand is the end-to-end effect of the fixed-budget refinement bundle, which jointly changes local step size, re-centering frequency, actor compute, and target/deployment dataflow.

The analyses test whether this bundle reduces large-budget low-return outcomes, whether the effect persists under the linearized realization, how first-hop target-policy displacement differs from final-policy reach, and which critic-validity diagnostics characterize failed training trajectories.

5.1 Fixed-budget splitting broadens the stability envelope

Figure 2 and Table 1 summarize the fixed-budget sweep. We descriptively call a high-budget cell collapsed when its two-seed mean score is below 20; this is a locomotion failure indicator, not a theoretical threshold. MPI-Prox with $K = 3$ has the strongest high-budget average and reduces the seed-mean collapse count from 35/63 to 17/63. MPI-Lin with $K = 3$ reduces the same count to 24/63, compared with 29/63 for two linearized hops, and raises the high-budget average from 36.68 to 43.91. The raw-seed counts confirm the lower failure rate of both three-hop realizations. The proximal $K = 2$ to $K = 3$ mean improvement for $T \geq 4$ is +10.76, whereas the median is only +0.50; the corresponding linearized differences are +7.23 and +0.86. Thus additional hops primarily rescue low-return outcomes rather than uniformly shifting settings. Appendix C reports sensitivity to thresholds 0, 10, 20, 30, and 40.

Task-dependent budget response. Environment-level curves exhibit recurring response motifs rather than a scalar task-specific optimum. HalfCheetah medium and replay are comparatively smooth, Hopper-replay has a wide high-return plateau, Hopper-medium and Walker2d-medium show sharp, nonmonotone transitions, and expert datasets show asymmetric high-budget failure. These motifs are descriptive, not disjoint task classes. In the matched Prox-2 audit, the current-state critic-coupled displacement proxy organizes many of these transitions more directly than nominal T (Appendix Figure 4). HalfCheetah-medium remains stable as $\hat{D}_{\text{critic}}$ grows from .033 to .056, whereas Hopper-expert falls from 101.9 to 1.8 as the proxy grows from .052 to .907. At comparable proxy values near .1, however, returns still differ substantially across tasks. Nominal budget is therefore an indirect control mapped through a task- and trajectory-dependent actor path and critic evolution. Reversals and cross-task dispersion prevent these curves from being summarized by a universal critic-valid radius; we use realized movement only to organize within-task transitions.

Targeted $K = 4$ extension. A separate-machine, four-seed MPI-Prox run on Hopper-medium shifts the high-return window to larger budgets. At $T = 17$, the $K = 3$ two-seed mean is 0.7, whereas $K = 4$ reaches 100.9 ± 1.7 . The $K = 4$ means at $T = \{10, 12, 14, 17\}$ are 100.4, 101.2, 101.1, and 100.9, with standard deviations at most 1.7. At $T = 20$, however, the four scores 99.7, 51.0, 0.7, and 15.8 yield 41.8 ± 44.0 ; at $T = 7$, $K = 4$ scores 84.8 ± 8.0 versus 94.4 for $K = 3$. These per-seed results support a K -dependent, nonmonotone stability window with a sharp variance increase near $T/K = 5$, not a verified $T_{\text{frontier}} \propto K$ law. Changing K jointly changes T/K , re-centering, compute, normalization, and routing, so this result does not identify a local-step mechanism (Appendix Table 8). The per-seed scores have been transferred, but the external raw evaluation files, configurations, and checkpoints have not; we therefore exclude this targeted run from the complete-grid aggregate.

The completed linearized sweep shows that the stability gain survives linear-target regression and is therefore not unique to the proximal actor objective. Three linearized hops exceed TD3+BC by 17.94 high-budget points and reduce collapse cells from 35 to 24; relative to two linearized hops, the average gain is 7.23 points and the collapse count falls from 29 to 24. MPI-Prox $K = 3$ remains stronger: over 63 paired high-budget cells, MPI-Lin $K = 3$ wins by more than one point in 8, 21 are within one point, and MPI-Prox wins in 34; the mean and median Lin-minus-Prox differences are -8.87 and -1.69 . Because the actor objectives differ, these remain comparisons of complete training procedures rather than isolated integrator effects.

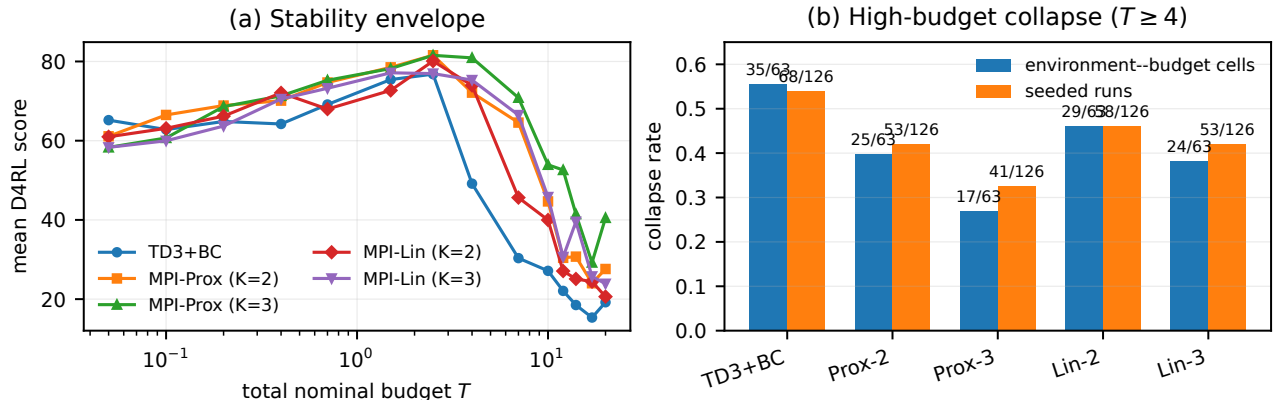


Figure 2: Fixed-budget sweep over nine locomotion tasks. (a) Mean score over tasks. (b) Collapse rates for $T \geq 4$, reported both by environment-budget cell and by seeded run. Fixed-budget substepping broadens the stable high-budget region for both proximal and scale-matched linearized realizations.

5.2 Large-budget behavior is nonlocal and critic dependent

At $T = 20$, the raw linearized target reaches the action box in 94–95% of states, and the effective action-step coefficient h_{eff} is only 14% and 9% of the nominal dT/K coefficient in the two hops. These are medians over 18 environment-seed runs per hop. The practical operation is projected target generation followed by amortized neural regression under a moving critic. The sweep measures the resulting end-to-end stability envelope; local truncation error lies outside this estimand.

Figure 3 shows the common failure signature. Environment return supplies the external label: collapsed runs score below 20. These runs have orders-of-magnitude larger TD-error tails, larger critic magnitudes, and larger realized movements. The controlled common critic is the frozen final TD3+BC $T = 1$ critic matched by environment and seed. A run-total gain sums the mean Q change across actor hops on the shared state batch. For collapsed MPI-Lin runs, its median under the jointly trained critic is 3.33×10^{11} ; under the common critic it is -1.42 , compared with $+2.11$ for stable runs. For MPI-Prox, the common-critic gain changes from $+2.51$ in stable runs to -1.06 in collapsed runs. This sign reversal diagnoses cross-critic disagreement consistent with Eq. (15); environment return remains the criterion for policy failure.

An exploratory CPU audit supplies a simulator reference for critic calibration. Across 60 MPI-Prox checkpoints, the median signed error $\hat{Q}_1(s, a_K) - Q_{\text{MC}}^1(s, a_K)$ is $+1.70 \times 10^{12}$ in five collapsed cells and -38.7 in 55 stable cells, corroborating severe overestimation in the collapsed screening cells. The action-ranking subset is sparse: only 69 of 720 cloned states have both action differences outside a one-return-unit dead zone, and 29 of those signs disagree. We therefore use this audit as checkpoint-level calibration evidence rather than as a routing or action-ranking estimand.

5.3 Target-policy displacement and final-policy reach

At the same nominal budget, MPI-Prox realizes a smaller current-state first-hop displacement proxy than MPI-Lin once $T \geq 7$ (Appendix Figure 5a). At $T = 20$, their median $\hat{D}_{\text{critic}}$ values are 0.195 and 0.323. Across 90 paired seeded comparisons at $T \in \{4, 7, 10, 14, 20\}$, exactly one realization collapses in 11 cases. In all 11, the stable realization has both smaller $\hat{D}_{\text{critic}}$ and smaller $\hat{D}_{\text{final}}$. These paired outcomes associate smaller realized movement with the stable optimization trajectory. Because the proxy is measured on current validation states rather than the Polyak target policy on next states, this result organizes the observed failure transition without identifying a causal route through the TD target.

We directly measure Eq. (13) on all 270 audited cells. The current-state proxy closely tracks deterministic target-policy displacement ($r = .9999$). MPI-Prox $K = 3$ has lower target-policy displacement than TD3+BC in 88 of 90 paired runs (median ratio .658), while the current-state $\hat{D}_{\text{final}}$ proxy is lower in only 36 of 90 (median ratio 1.057). Thus the three-hop refinement can reduce movement in the target-policy branch without merely shrinking its measured final-policy reach. Against $K = 2$, $K = 3$ lowers target-policy displacement in 84 of 90 runs but loses by more than one score point in 17 of those 84, showing that target-policy displacement alone does not determine performance (Appendix Table 10).

A targeted $K = 3$ study on Hopper-medium reveals two modes of decoupling (Appendix Figure 5b). For $T = 4$ – 10 , $K = 2$ and $K = 3$ have nearly identical $\hat{D}_{\text{critic}}$, whereas $K = 3$ increases $\hat{D}_{\text{final}}$ and score; at $T = 10$, the scores are 58.12 and 85.35. In the observed transition region from stable to collapsed scores ($T = 12, 14$), $K = 3$ instead contracts both displacements and avoids the low-return outcome. At $T = 20$, both enter severe projection and saturation and collapse. The third hop changes target-policy movement,

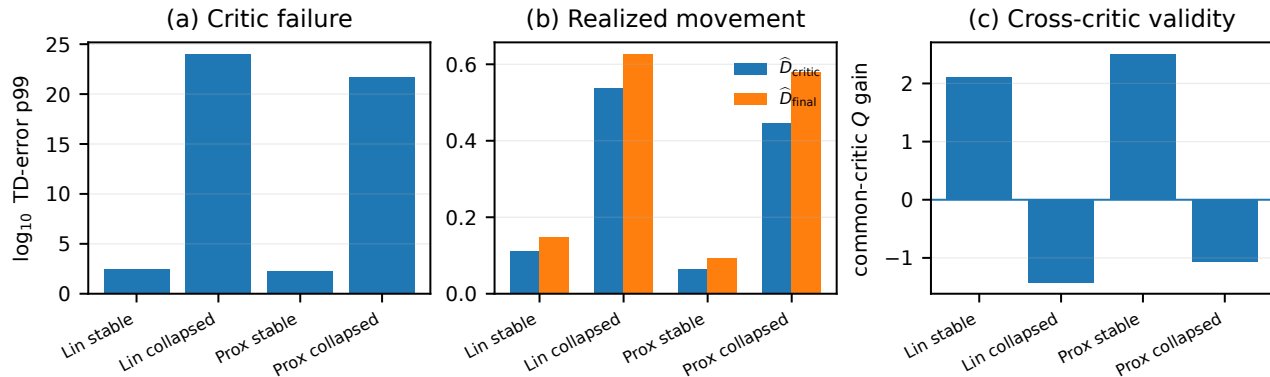


Figure 3: Median diagnostics over the matched $T \in \{4, 7, 10, 14, 20\}$ audit. Proximal and linearized realizations share a diagnostic pattern consistent with critic miscalibration: critic errors and realized movements increase sharply, and run-total gains under a common critic become negative. Environment return labels failure; the common critic diagnoses cross-critic disagreement.

re-centering frequency, and the optimization trajectory rather than the formal order of the local update.

5.4 Targeted component controls

We next hold the actor trajectory fixed and change only the backup route. Two actor-detached shadow critics start from the same parameters and receive identical minibatches and target-smoothing noise; one bootstraps through the first-hop target actor and the other through the final-hop target actor. The study covers Hopper-medium at $T \in \{10, 20\}$ and Walker2d-medium at $T \in \{4, 7\}$, with two seeds and four checkpoints. Relative to a common simulator return under the effective first-route backup continuation, the final-route shadow has greater state-median symmetric relative error in all eight run-level averages. The final-minus-first route delta has mean .124, median .013, a run-bootstrap mean 95% interval [.008, .323], and an exact two-sided sign-flip $p = .0078$. Because the final-route critic targets a different continuation, this contrast measures route-induced alignment with the primary first-route value estimand, not each shadow critic’s calibration to its own continuation. Neither shadow critic drives the actors, so the intervention does not estimate a routing effect on return.

Two additional eight-run controls probe simpler explanations (Appendix Table 9). With three actor optimizer calls in every condition, sequential $K = 3$, a fixed-reference chain, and a direct-final control have means 71.4, 70.0, and 59.4 and collapse counts 2/8, 2/8, and 3/8. Fixed-reference refinement therefore matches sequential re-centering in this targeted sample, so the experiment does not establish re-centering alone as the source of the sweep-wide gain. Slowing only the $K = 1$ actor-target Polyak rate by a factor of three changes the matched collapse count from 6/8 to 7/8 and does not reproduce the $K = 3$ rescue; actual displacement under this lag intervention was not audited. Together, the controls support first-hop backup routing while retaining the implemented

bundle, rather than any isolated component, as the main estimand.

5.5 Discussion and limitations

Both three-hop realizations reduce large-budget failures, although MPI-Prox remains stronger overall. Direct measurements show that lower first-hop target movement need not shrink final-policy reach, and fixed-trajectory final-route shadows are less aligned with the common first-route value estimand in all eight targeted runs. Fixed-reference refinement matches sequential re-centering, while slower target averaging does not rescue the selected one-hop cells. Thus the primary estimate remains the implemented bundle: K changes local step, re-centering, compute, normalization, and routing together. The fourth-hop study likewise relocates rather than uniformly improves the high-return window.

The complete grid covers TD3+BC on nine locomotion datasets with two paired seeds per cell. The separate-machine four-seed $K = 4$ study lacks transferred raw artifacts. Collapse is descriptive, and the common-critic, simulator, and eight-run controls are targeted rather than exhaustive. The finite-step margin concerns the learned critic; the Wasserstein-flow account assumes small realized steps under a fixed critic.

6 CONCLUSION

MPI separates critic-target and deployment roles across a re-centered TD3+BC actor chain, with a deterministic Wasserstein proximal interpretation in the fixed-critic local regime. In the complete sweep, three proximal and linearized hops reduce large-budget low-return cells from 35/63 to 17/63 and 24/63, respectively. Three-hop MPI lowers target-policy displacement in 88/90 pairs without generally shrinking the final-policy proxy, and the first-route shadow is closer to the common first-route simulator estimand in all eight targeted runs. These results support the end-to-

end architectural claim that fixed-budget substepping broadens empirical stability while separating target exposure from deployment reach; they do not attribute the sweep-wide gain to routing, re-centering, or compute alone.

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A PROOFS AND ADDITIONAL ANALYSIS

A.1 Behavior-anchor decomposition and Proposition 1

For any square-integrable random action a conditioned on s , let $b(s) = \mathbb{E}[a \mid s]$. The bias-variance identity

gives

$$\mathbb{E}[\|\mu(s) - a\|_2^2 \mid s] = \|\mu(s) - b(s)\|_2^2 + \mathbb{E}[\|a - b(s)\|_2^2 \mid s]. \quad (21)$$

Averaging over states and dividing by d proves Eq. (7); the second term is independent of μ .

For state-conditioned deterministic policies $\pi_\mu(\cdot \mid s) = \delta_{\mu(s)}$ and $\pi_\nu(\cdot \mid s) = \delta_{\nu(s)}$,

$$W_2^2(\pi_\mu(\cdot \mid s), \pi_\nu(\cdot \mid s)) = \|\mu(s) - \nu(s)\|_2^2. \quad (22)$$

Consequently, d_d^2 is a fixed rescaling of the integrated squared Wasserstein distance. The normalized anchored objective is

$$-\alpha \mathbb{E}_s \bar{Q}(s, \mu(s)) + d_d^2(\mu, \nu). \quad (23)$$

Multiplication by the positive constant $1/\alpha$ identifies it with Eq. (8) when $1/(2h) = 1/\alpha$, or $h = \alpha/2$. If the value term is instead $-\lambda \mathbb{E}_s \bar{Q}$, then $\lambda \hat{Q} = \lambda C \bar{Q}$ and $h = \lambda C/2$.

A.2 Proof of Proposition 2

By ε_k -optimality relative to the feasible comparator μ_{k-1} ,

$$\begin{aligned} & -\frac{2h_k}{C_k} \mathbb{E}_s \hat{Q}(s, \mu_k(s)) + d_d^2(\mu_k, \mu_{k-1}) \\ & \leq -\frac{2h_k}{C_k} \mathbb{E}_s \hat{Q}(s, \mu_{k-1}(s)) + \varepsilon_k. \end{aligned} \quad (24)$$

Rearranging yields Eq. (14).

A.3 Critic-error transfer

Let $\hat{Q} = Q^\dagger + e$. Substitution into Eq. (14) gives

$$\begin{aligned} m_k^{\hat{Q}} &= \frac{2h_k}{C_k} \mathbb{E}_s [Q^\dagger(\mu_k) - Q^\dagger(\mu_{k-1})] - d_d^2(\mu_k, \mu_{k-1}) \\ &+ \frac{2h_k}{C_k} \mathbb{E}_s [e(\mu_k) - e(\mu_{k-1})] \\ &= m_k^{Q^\dagger} + \frac{2h_k}{C_k} \mathbb{E}_s [e(\mu_k) - e(\mu_{k-1})]. \end{aligned} \quad (25)$$

This proves Eq. (15). If $|e(s, a)| \leq \delta_k$ on the realized path, then

$$m_k^{Q^\dagger} \geq -\varepsilon_k - \frac{4h_k}{C_k} \delta_k. \quad (26)$$

The result concerns a trusted critic objective. A true-return statement would require additional occupancy and value-estimation assumptions.

A.4 Proof of Proposition 3

Let $F_{s'}(a) = \min_j Q_j^-(s', a)$. If every $Q_j^-(s', \cdot)$ is $L(s')$ -Lipschitz, then

$$\begin{aligned} |F_{s'}(a) - F_{s'}(b)| &\leq \max_j |Q_j^-(s', a) - Q_j^-(s', b)| \\ &\leq L(s') \|a - b\|_2. \end{aligned} \quad (27)$$

Therefore,

$$|y_{\mu_1} - y_{\mu_0}| \leq \gamma L(s') \|\mu_1(s') - \mu_0(s')\|_2. \quad (28)$$

Squaring, averaging over dataset transitions, and taking the square root proves Eq. (16). Later actors do not occur because the TD target conditions only on μ_1 .

A.5 Local explicit-implicit equivalence

Let $\beta = dh/C$, $g = \nabla_a \hat{Q}(a_0)$, and suppose the action constraint is inactive. The scale-matched linearized update is $\Delta a_{\text{lin}} = \beta g$. A statewise exact proximal solution satisfies

$$\Delta a_{\text{prox}} = \beta \nabla_a \hat{Q}(a_0 + \Delta a_{\text{prox}}). \quad (29)$$

If $\hat{Q}$ is twice continuously differentiable near a_0 ,

$$\nabla_a \hat{Q}(a_0 + \Delta a) = g + H \Delta a + O(\|\Delta a\|^2). \quad (30)$$

Thus

$$(I - \beta H) \Delta a_{\text{prox}} = \beta g + O(\beta \|\Delta a_{\text{prox}}\|^2). \quad (31)$$

For $\beta \|H\| < 1$ and sufficiently small β ,

$$\Delta a_{\text{prox}} = \beta g + \beta^2 H g + O(\beta^3), \quad (32)$$

whereas $\Delta a_{\text{lin}} = \beta g$. Both are first-order schemes for fixed positive C , and changing a fixed C only rescales time. A positive policy-dependent global C preserves the raw-critic ascent direction under an adaptive time parameterization in the infinitesimal limit. This derivation is not applied to the large-budget regime with active projection, ReLU-region crossings, critic drift, or inaccurate actor solves.

A.6 Proof of Proposition 4

Write $x_j = x_0 + j h f_0 + O(h^2)$ for fixed K , which follows inductively from the one-step consistency assumption. Smoothness then gives

$$f(x_j) = f_0 + j h J_0 f_0 + O(h^2). \quad (33)$$

Summing the K increments,

$$\begin{aligned} x_K - x_0 &= h \sum_{j=0}^{K-1} f(x_j) + h^2 \sum_{j=0}^{K-1} p(x_j) + O(Kh^3) \\ &= K h f_0 + h^2 \frac{K(K-1)}{2} J_0 f_0 \\ &\quad + K h^2 p_0 + O(h^3). \end{aligned} \quad (34)$$

where the final remainder uses fixed K . Substituting $h = T/K$ proves Eq. (19). The exact flow satisfies $x(T) = x_0 + T f_0 + \frac{1}{2} T^2 J_0 f_0 + O(T^3)$. Thus explicit Euler has leading error $-T^2 J_0 f_0 / (2K)$, while backward Euler has $+T^2 J_0 f_0 / (2K)$. Both remain first-order methods.

B EXPERIMENTAL DETAILS

B.1 Algorithms and schedules

The extension table reports the original CORL-based MPI evaluation of IQL, ReBRAC, and TD3+BC with one or two equal-step refinements. Per-algorithm launch grids, configurations, and model-selection meta-data are not present in the current artifact. We therefore treat the table as retrospective evidence of refinement capacity, not as a selection-free benchmark comparison.

For the fixed-budget TD3+BC study, the total normalized budget is split as $h = T/K$. MPI-Prox uses Eq. (9) with a moving reference; MPI-Lin constructs Eq. (20) and uses MSE regression. The K actor networks share an architecture and learning rate but are independently initialized, persistent networks with separate Adam states. At every second critic update, the implementation performs one optimizer step per actor in chain order on the same minibatch. The base actor is updated first; then its actor target and the critic target receive a Polyak update of 0.005 before later actors use stop-gradient references from the updated online chain. The scale C_k is recomputed from each hop’s reference actions. All methods therefore match data, critic architecture, and 10^6 critic updates, but a K -hop run performs K actor optimizer steps and is not compute matched. We use batch size 256, learning rate 3×10^{-4} , discount 0.99, and ten evaluation episodes every 5000 updates. The 14-value grid spans 0.05 to 20; the complete sweep uses two paired seeds and reports both cell-level and raw-seed failure counts.

B.2 Diagnostic quantities

On a fixed current-state validation batch with paired dataset actions, we compute the empirical action-averaged proxies

$$\begin{aligned}\hat{D}_{\text{critic}} &= \mathbb{E}_{(s,a) \sim \mathcal{D}} \frac{1}{d} \|\mu_1(s) - a\|_2^2, \\ \hat{D}_{\text{final}} &= \mathbb{E}_{(s,a) \sim \mathcal{D}} \frac{1}{d} \|\mu_K(s) - a\|_2^2.\end{aligned}$$

These quantities measure the online actor chain relative to its sample anchor. They are proxies for, rather than direct measurements of, the next-state target-policy displacement in Proposition 3: the practical TD target uses a Polyak target actor and policy-smoothing noise. By Eq. (7), the population expectation of each proxy equals squared displacement from the conditional-mean behavior map plus the same additive conditional-action variance. Population differences therefore cancel this common floor, but ratios do not: a ratio of sample-anchored proxies is not a ratio of mean-policy displacements. Finite validation batches add sampling error to both interpretations. We additionally evaluate the deterministic Polyak target actor and 64 draws of clipped target-smoothing noise on 4096 next-state transitions per checkpoint. All target-exposure quantities remain squared displacement per

action coordinate relative to the paired next dataset action. Thus absolute levels include conditional action variance, whereas paired differences use a common sampled anchor. For MPI-Lin, projection attenuation is the RMS norm ratio between the clipped and raw target displacements. If $\tilde{g}$ is the normalized critic action gradient and Δa is the learned actor displacement, define

$$h_{\text{eff}} := \frac{\mathbb{E} \langle \Delta a, \tilde{g} \rangle}{\mathbb{E} \|\tilde{g}\|_2^2}.$$

We report $h_{\text{eff}}/(dT/K)$ as the effective fraction of the nominal linear-target coefficient. The matched audit evaluates 512 fixed states per environment–seed run. We also record action saturation, TD-error quantiles, critic magnitude, critic loss, and gains under both the jointly trained critic and a controlled common critic: the frozen final TD3+BC $T = 1$ critic matched by environment and seed. A run-total gain is the sum of mean Q changes across the K hops on this shared batch. The common critic supports cross-critic comparison but is not treated as ground truth. A run is descriptively labeled collapsed only from its external normalized return, using the threshold 20.

For the exploratory simulator-reference audit, we clone MuJoCo integration state and environment RNG from evaluation-policy trajectories, record the episode clock, and reset the TimeLimit for a continuing-task critic-calibration estimand. At each state we compare $a_1 = \mu_1(s)$ and $a_K = \mu_K(s)$ under continuations ρ_1 , μ_1 , and μ_K ; the primary calibration target here is the discounted return under the effective backup continuation ρ_1 . The CPU screening covers three environments, $T \in \{4, 7, 10, 14, 20\}$, $K \in \{2, 3\}$, two seeds, and 12 states per checkpoint. It uses a 1000-step horizon, $\gamma = .99$, four common-random-number target-smoothing rollouts, and a one-return-unit ranking dead zone. It is a post-hoc screening study, not a causal routing intervention.

The route-only intervention trains two actor-detached shadow critics from identical initial parameters on the same minibatches and target-smoothing draws while the primary first-hop critic continues to drive the shared actor trajectory. One shadow uses the first-hop target actor and the other the final-hop target actor. The simulator audit evaluates 12 cloned states at four checkpoints in each of eight runs; inference aggregates checkpoint-level state medians to the training-run level.

The compute-matched study uses the same eight final cells. Sequential $K = 3$ performs actor updates anchored at μ_0, μ_1 , and μ_2 ; the fixed-reference control anchors both later actors at μ_1 ; the direct-final control applies two optimizer calls to one $2T/3$ objective anchored at μ_1 . Every condition performs three actor optimizer calls per delayed update. The target-lag control instead keeps $K = 1$ and changes only the actor-target Polyak rate from .005 to .005/3, leaving the critic-target rate unchanged.

The frozen-critic small-step audit uses the final TD3+BC $T = 1$ critic for all nine environments and two seeds. At 512 interior dataset actions per run it compares the projected explicit step with a projected fixed-point solution of the implicit action equation for $h \in \{.001, .003, .01, .03, .1\}$. This is an action-level local-consistency test, not a return or coupled-training experiment.

C ADDITIONAL RESULTS

Table 8: Targeted Hopper-medium scores. The $K \leq 3$ entries are two-seed means from the complete sweep; Prox-4 reports four-seed mean $\pm$ sample standard deviation from a separate-machine run and is used only for the stability-window comparison.

T	TD3+BC	Prox-2	Prox-3	Prox-4
7	48.1	83.2	94.4	84.8 $\pm$ 8.0
10	0.7	50.8	99.5	100.4 $\pm$ 1.1
12	0.9	0.7	93.6	101.2 $\pm$ 0.4
14	0.7	50.6	88.1	101.1 $\pm$ 0.7
17	0.7	0.6	0.7	100.9 $\pm$ 1.7
20	0.7	0.7	51.5	41.8 $\pm$ 44.0

Table 9: Targeted final-return controls on the same eight environment–budget–seed cells. The three $K = 3$ rows are compute matched to three actor optimizer calls; the $K = 1$ rows differ only in actor-target Polyak rate.

Condition	Mean	Median	Score < 20
Sequential $K = 3$	71.4	88.9	2/8
Fixed-reference $K = 3$	70.0	87.9	2/8
Direct-final $K = 3$	59.4	89.6	3/8
Canonical $K = 1$	13.6	0.8	6/8
3 $\times$ actor-target lag, $K = 1$	12.0	1.5	7/8

The paired route-only study uses the same eight runs but aggregates four checkpoints per run. For symmetric relative error against the common first-route simulator estimand, all eight run-level final-route minus first-route differences are positive. Their mean is .124, median is .013, exact two-sided sign-flip $p = .0078$, and run-bootstrap mean 95% interval is [.008, .323]. Because the routes imply different continuation policies, this estimates their relative alignment with the primary first-route value estimand rather than calibrating each shadow to its own continuation; it is not a return-effect estimate.

In the frozen-critic audit, 15 of 18 environment–seed runs have at least one converged, projection-free point eligible for the local comparison. Every eligible run has maximum local relative explicit–implicit discrepancy below 2.7%; the median run maximum is .55%. Three runs have no eligible point, and a log–log slope is identifiable in only one run because a frozen ReLU critic can produce floating-point agreement inside one activation region. The audit supports local numerical agreement but does not estimate a universal $O(h^2)$ rate.

C.1 Direct target exposure and simulator-reference screening

The direct target-policy audit covers 270 checkpoints: three methods, nine environments, five budgets, and two seeds. Table 10 separates critic-target exposure from final-policy reach. The ratios are descriptive ratios of sample-anchored squared displacements and are not ratios of conditional-mean policy distances.

Table 10: Paired CPU audit of deterministic next-state Polyak-target exposure and current-state final-policy displacement. The Lower column counts runs in which the left method has the smaller quantity.

Comparison	Quantity	Lower	Median ratio
Prox-3 / TD3	Target exposure	88/90	.658
Prox-3 / TD3	Final displacement	36/90	1.057
Prox-3 / Prox-2	Target exposure	84/90	.903

The current-state proxy and deterministic target exposure have Pearson correlation .9999 over all 270 cells; their median ratio is .997 and the largest cellwise relative difference is 5.7%. The corresponding correlation after target smoothing is .9993. The close alignment is an empirical property of these checkpoints, where Polyak lag is small, rather than a mathematical equivalence. Among the 84 Prox-3/Prox-2 runs in which Prox-3 lowers deterministic target exposure, Prox-3 is nevertheless worse by more than one score point in 17. This rejects the stronger interpretation that minimizing exposure alone determines performance.

In the simulator-reference screening, 55 stable cells contribute 660 states and five collapsed cells contribute 60. At a_K , the median signed online critic error relative to $Q_{MC}^{p_1}$ is -38.7 in stable cells and $+1.70 \times 10^{12}$ in collapsed cells. This independently corroborates the overestimation signature in Figure 3. It does not yet provide a reliable action-ranking rate: only 69 of 720 states have both action differences outside the dead zone, with 29 sign disagreements, and within-checkpoint states are correlated. Submission-level use requires checkpoint-level uncertainty and a larger pre-specified state sample.

D REPRODUCIBILITY CHECKLIST

1. Models and algorithms. The mathematical setting, normalized actor objective, re-centered update, scale-matched control, and critic coupling are specified in Sections 4–5. Base configurations follow the cited CORL implementations. Exact launchers and configurations are included for the TD3+BC fixed-budget study; original extension launch grids, configurations, and selection meta-data are not archived.
2. Theoretical claims. Assumptions and proofs for the operator identification, finite-step margin, critic-error transfer, bootstrap-exposure bound, and local limit are provided in the main text and Appendix A. No true-return monotonicity is

Table 2: Five-seed extension results from the original MPI evaluation. Values are mean $\pm$ standard deviation.

Environment	Base	IQL +1	+2	Base	ReBRAC +1	+2	Base	TD3+BC +1	+2
HalfCheetah-M	48.42 $\pm$.19	53.78 $\pm$.24	55.94 $\pm$.67	64.90 $\pm$.82	65.92 $\pm$ 1.51	66.00 $\pm$ 1.58	48.40 $\pm$.38	54.46 $\pm$.36	57.66 $\pm$.51
HalfCheetah-MR	43.84 $\pm$.88	49.80 $\pm$.50	52.30 $\pm$.87	50.62 $\pm$ 2.26	50.73 $\pm$.95	49.98 $\pm$.49	44.70 $\pm$.68	49.76 $\pm$.83	51.98 $\pm$ 1.06
HalfCheetah-ME	90.44 $\pm$ 4.06	91.02 $\pm$ 3.72	88.58 $\pm$ 5.86	103.00 $\pm$ 4.76	97.12 $\pm$ 4.34	97.08 $\pm$ 2.04	90.62 $\pm$ 6.51	89.88 $\pm$ 3.72	74.42 $\pm$ 8.52
Hopper-M	60.44 $\pm$ 6.17	90.54 $\pm$ 9.40	101.40 $\pm$ 3.51	98.35 $\pm$ 5.09	101.05 $\pm$ 1.49	101.22 $\pm$ 2.29	59.18 $\pm$ 3.56	91.44 $\pm$ 7.43	100.60 $\pm$ 1.54
Hopper-MR	86.72 $\pm$ 7.58	100.50 $\pm$ 3.36	103.04 $\pm$.42	90.28 $\pm$ 20.12	100.70 $\pm$.29	99.53 $\pm$ 1.54	71.02 $\pm$ 23.88	99.32 $\pm$ 1.49	99.12 $\pm$ 1.64
Hopper-ME	102.20 $\pm$ 21.41	106.14 $\pm$ 10.82	107.38 $\pm$ 10.84	111.65 $\pm$.45	109.65 $\pm$ 2.73	107.22 $\pm$ 7.96	94.62 $\pm$ 9.48	96.82 $\pm$ 12.32	105.48 $\pm$ 8.26
Walker2d-M	80.54 $\pm$ 3.57	74.94 $\pm$ 15.15	85.26 $\pm$ 3.34	85.75 $\pm$.64	85.55 $\pm$ 1.05	86.97 $\pm$.97	83.78 $\pm$ 2.77	84.30 $\pm$ 6.49	89.44 $\pm$ 2.10
Walker2d-MR	78.06 $\pm$ 8.59	81.82 $\pm$ 3.39	84.46 $\pm$ 5.58	79.55 $\pm$ 5.40	86.33 $\pm$ 5.04	88.35 $\pm$ 3.73	79.30 $\pm$ 6.07	85.66 $\pm$ 2.55	89.42 $\pm$ 1.89
Walker2d-ME	111.86 $\pm$ 1.06	112.62 $\pm$.77	113.78 $\pm$ 1.22	111.83 $\pm$.28	112.08 $\pm$.79	113.38 $\pm$.24	110.24 $\pm$.18	112.20 $\pm$.64	113.62 $\pm$ 1.50
Average	78.06	84.57	88.02	88.44	89.90	89.97	75.76	84.87	86.86

Table 3: Mean score over nine tasks at each total budget in the fixed-budget sweep.

T	.05	.1	.2	.4	.7	1.5	2.5	4	7	10	12	14	17	20
TD3+BC	65.19	62.81	64.85	64.22	69.14	75.43	76.80	49.16	30.35	27.14	22.08	18.51	15.35	19.21
Prox-2	61.11	66.50	68.87	70.11	74.69	78.52	81.57	72.13	64.57	44.67	30.43	30.73	24.01	27.61
Prox-3	58.31	60.69	68.61	71.32	75.25	78.21	81.55	80.91	70.84	53.91	52.60	41.47	29.23	40.51
Lin-2	60.98	63.12	66.20	72.07	67.98	72.70	80.10	73.95	45.63	39.98	27.09	25.10	24.41	20.63
Lin-3	58.32	59.97	63.69	70.52	73.23	77.14	76.94	75.31	66.47	45.78	30.65	39.54	25.74	23.88

Table 4: Collapse-count sensitivity for $T \geq 4$. Entries are seed-mean cells (out of 63) / raw seeded runs (out of 126). Prox-3 has the fewest failures for thresholds 10–40; Lin-3 improves on Lin-2 throughout this range. Threshold 0 isolates negative scores.

Threshold	TD3	Prox-2	Prox-3	Lin-2	Lin-3
0	2/12	2/6	1/6	0/6	2/12
10	29/66	20/51	15/39	28/54	21/46
20	35/68	25/53	17/41	29/58	24/53
30	36/74	26/56	19/44	29/63	24/55
40	39/83	26/60	20/48	32/64	24/57

Table 5: Per-environment high-budget ($T \geq 4$) means.

Environment	TD3	Prox-2	Prox-3	Lin-2	Lin-3
Hopper-M	16.0	35.2	69.4	25.5	46.3
Hopper-MR	41.4	90.6	99.7	90.5	99.4
Hopper-ME	3.4	17.5	30.7	16.6	27.7
HalfCheetah-M	59.0	59.1	59.2	53.4	55.8
HalfCheetah-MR	51.7	52.6	51.6	46.5	49.2
HalfCheetah-ME	18.4	24.5	28.5	18.8	22.4
Walker2d-M	10.3	24.3	34.3	9.3	26.5
Walker2d-MR	31.3	49.0	76.6	51.2	43.5
Walker2d-ME	2.2	25.4	25.0	18.5	24.4

claimed.

- Empirical results. The number of tasks, budgets, seeds, collapse definition, threshold sensitivity, and aggregation rules are reported. Retrospective extension results use five seeds; the complete proximal and linearized grids use two paired seeds and report raw-seed as well as seed-mean failure counts. The targeted Prox-4 comparison uses four transferred per-seed scores from a separate machine and is excluded from the complete-grid aggregate. Route, compute, lag, and frozen-critic studies report their targeted inference units and limitations.
- Code and assets. Experiments use public D4RL datasets and CORL baseline implementations. The accompanying anonymized package covers the reported TD3+BC fixed-budget sweeps, controls, and diagnostic scripts. The external Prox-4 score matrices are available, but their raw evaluation files, configurations, and checkpoints have not been transferred; complete reproduction and selection auditing of the original extension analysis is also

Table 6: Matched $K = 2$ diagnostic medians over $T \in \{4, 7, 10, 14, 20\}$.

	Lin st.	Lin col.	Prox st.	Prox col.
No. runs	52	38	59	31
TD p99	270	9.6×10^{23}	160.8	5.0×10^{21}
$ Q $	311	8.5×10^{11}	346.7	6.9×10^{10}
Critic loss	34	1.2×10^{23}	18.7	8.8×10^{20}
$\hat{D}_{\text{critic}}$	.112	.538	.064	.446
$\hat{D}_{\text{final}}$	.147	.626	.094	.578
Total common-critic gain	+2.11	−1.42	+2.51	−1.06

unavailable.

- Human subjects and sensitive data. Not applicable.

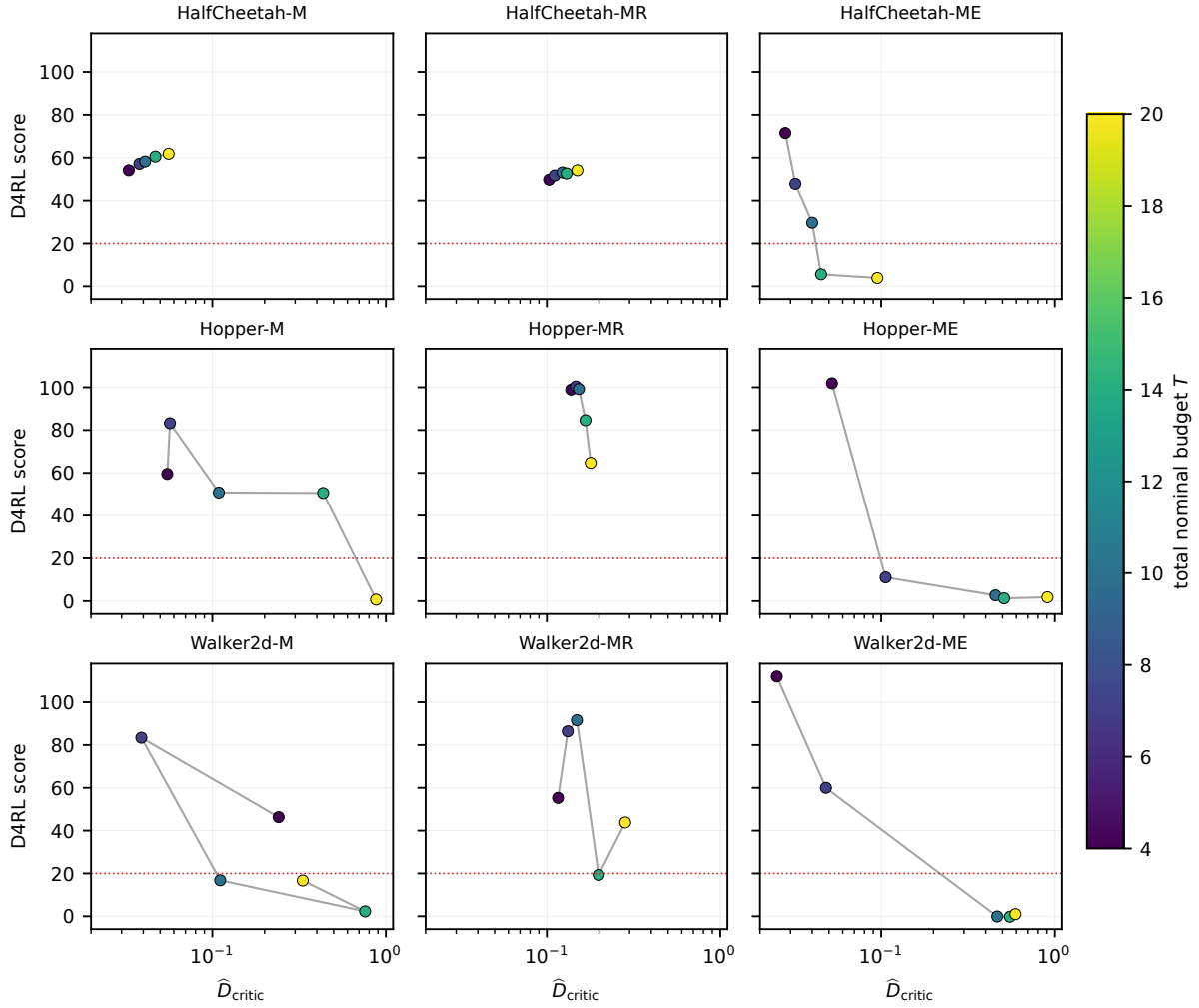


Figure 4: Environment-wise return versus the current-state $\hat{D}_{\text{critic}}$ proxy for MPI-Prox $K=2$. Points are two-seed means for the common-batch audit at $T \in \{4, 7, 10, 14, 20\}$; color denotes T . Lines connect increasing T , exposing reversals rather than imposing a monotone response. Shared axes show that realized movement organizes several within-task transitions but does not define a universal cross-task failure threshold. The proxy does not measure the next-state Polyak target-policy movement.

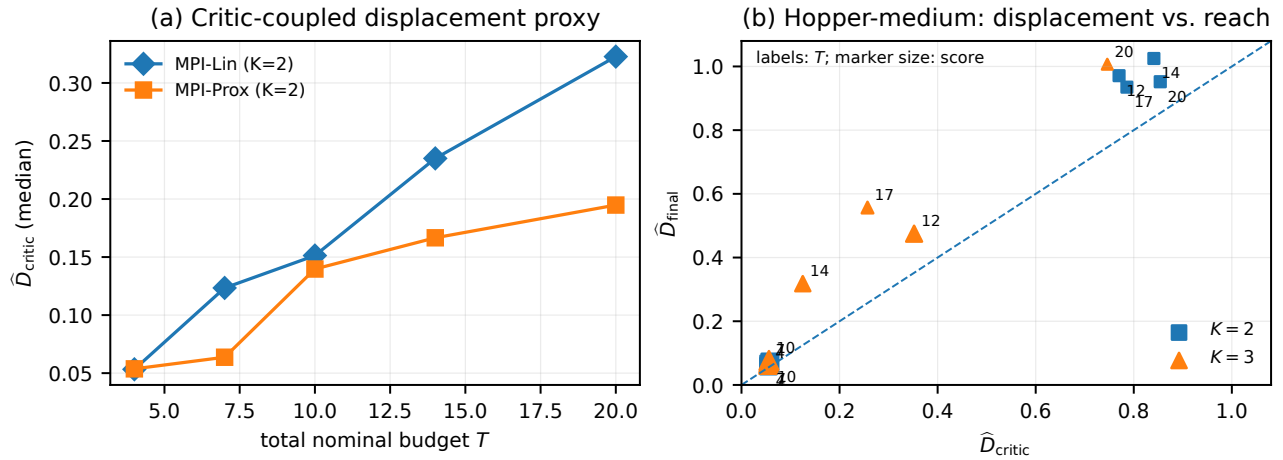


Figure 5: Realized movement. (a) The proximal realization produces a smaller current-state first-hop displacement proxy for $T \geq 7$. (b) On Hopper-medium, $K=3$ either increases final reach at approximately fixed displacement or contracts both displacements in the observed score-transition region. The displayed current-state quantities exclude Polyak target lag and smoothing noise. Labels indicate T and marker size is proportional to score.

Table 7: Targeted Hopper-medium comparison of scale-matched linearized $K = 2$ and $K = 3$. The displacement columns are medians over two seeds; R_{rms} is the median of the per-seed RMS ratio $\sqrt{\widehat{D}_f/\widehat{D}_c}$. Because both sample-anchored proxies contain an additive behavior-action variance floor, R_{rms} is descriptive and is not a population ratio of final to critic-coupled mean-policy displacement.

T	Score		$K = 2$			$K = 3$		
	Lin-2	Lin-3	$\widehat{D}_c$	$\widehat{D}_f$	R_{rms}	$\widehat{D}_c$	$\widehat{D}_f$	R_{rms}
4	53.45	64.66	.055	.059	1.040	.054	.062	1.063
7	63.36	74.26	.056	.065	1.075	.055	.069	1.115
10	58.12	85.35	.058	.072	1.113	.056	.077	1.168
12	0.66	50.53	.770	.971	1.123	.352	.475	1.168
14	0.66	41.20	.841	1.025	1.105	.125	.318	1.453
17	0.69	7.73	.786	.935	1.091	.257	.557	1.496
20	1.28	0.66	.854	.952	1.055	.746	1.007	1.162